MATH317 · NUMERICAL METHODS

# تدريب Midterm 461

2024 · 60 دقيقة · 25 درجة

**محارب**

## 01 الأسئلة

حل كل سؤال في المساحة اللي تحته. الحل الكامل في ملف منفصل.

**Q1** 6 + 5 M

**1. A program approximates square roots:**

x = 4: true √x = 2.0000, computed 2.1000.

x = 10: true √x = 3.1623, computed 3.1600.

a) Define the absolute and relative errors. b) Compute the absolute errors. c) Compute the relative errors as percentages.

**2. Given $f(x) = x^3 - 2x - 1$ and $x_0 = 1.5$, apply Newton's method for two iterations.**

6 M **Q2**

Determine the single-precision machine representation of the decimal number 1259.125.

2 + 6 M **Q3**

Given $f(x) = e^x - 4x^2$.

a) Prove that f(x) has exactly one real root in the interval [0, 1].

b) Find a root on [0, 1] by using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than $0.5 \times 10^{-6}$.